Course Code : ECT 456-2 **QTSR/RW – 24 / 1807**

Eighth Semester B. Tech. (Electronics and Communication Engineering) Examination

COMPUTER VISION

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) All questions are compulsory and carry marks as indicated.
(2) Assume suitable data wherever necessary.
(3) Illustrate your answers with the help of neat sketches.
(4) Use of Non – programmable calculator is permitted.

1. (a) Elaborate the various applications of computer vision and justify any one with example. 5(CO1)

(b) What is vanishing point, Explain and derive equations with the help of parallel lines and how to project vanishing point on 2D image plane. 5(CO1)

2. (a) Derive the equation of perspective imaging with pinhole and analyze the concept of camera obscura. 4(CO2)

(b) Derive the equation for determining image magnification using two lens system and elaborate the concept. 6(CO2)

3. (a) Discuss the types of image sensors with neat diagram, mentioning the advantages of each type. 6(CO1,2)

(b) Derive the equations for image formation in stereo vision in X,Y,Z coordinates with the help of diagram. 4(CO1,2)

4. (a) Elaborate edge detection in image and explain its importance in computer vision with neat diagram. 4(CO4)

(b) Derive the equations for gradient edge detector with kernal and elaborate important features of gradient edge detector. 6(CO4)

5. (a) Explain the concept of Linear Regression Process using Machine Learning with algorithm. 6(CO3,4)

   (b) How does decision function work on image formation using machine learning concepts, explain in brief. 4(CO4,5)

6. Elaborate the concept of pattern classification using Artificial neural Network with the help of algorithm. 10(CO5)